VPC Questions: SAA-C03 Level

1. What is the purpose of a VPC in AWS?

A. It provides a virtual network environment to launch AWS resources
B. It encrypts all network traffic by default
C. It manages IAM roles automatically
D. It provides DNS services only

2. Which component allows your VPC to connect to the internet?

A. NAT Gateway
B. Internet Gateway
C. VPC Peering
D. Route 53 Resolver

3. You want an EC2 instance in a private subnet to access the internet for software updates. What should you use?

A. Internet Gateway
B. VPC Peering
C. NAT Gateway
D. Local Gateway

4. What does a route table do?

A. Controls inbound/outbound traffic at instance level
B. Determines how traffic is directed within a VPC
C. Encrypts data in transit
D. Assigns IPs to new instances

5. Which is true about Security Groups?

A. They are stateless
B. They allow deny rules
C. They are stateful and allow only allow rules
D. They are attached to subnets

6. Network ACLs (NACLs) are…

A. Stateful firewalls attached to instances
B. Stateless firewalls attached to subnets
C. Used only for outbound traffic
D. Attached to IAM users

7. What happens when you create a new VPC?

A. It automatically creates public and private subnets
B. It automatically creates a route table, NACL, and security group
C. It comes with an Internet Gateway by default
D. It automatically enables VPC Flow Logs

8. How can two VPCs in the same region communicate privately without going over the internet?

A. NAT Gateway
B. Internet Gateway
C. VPC Peering
D. Route 53 Resolver

9. Which IP range is valid for a VPC CIDR block?

A. 200.168.0.0/16
B. 172.16.0.0/12
C. 256.1.1.0/24
D. 300.0.0.0/16

10. You want your on-premise network to securely connect to AWS VPC using IPSec. Which service is used?

A. Transit Gateway
B. Direct Connect
C. Site-to-Site VPN
D. VPC Peering

11. Your company has an EC2 instance in a private subnet that must securely access Amazon S3 without using a NAT Gateway. What should you configure?

A. Create a VPC Peering connection between the VPC and S3
B. Create a Gateway VPC Endpoint for S3
C. Attach an Internet Gateway to the VPC
D. Use an Interface VPC Endpoint for EC2

12. You have two VPCs (VPC-A and VPC-B) in the same region. EC2 in VPC-A cannot communicate with EC2 in VPC-B even after VPC peering. What is the most likely reason?

A. The VPC CIDR ranges overlap
B. The VPCs are in different AZs
C. Security Groups do not support cross-VPC rules
D. Interface endpoints block peering traffic

13. A database running in a private subnet needs to accept connections only from an application server in a public subnet. Which configuration ensures this?

A. Add inbound rules in the NACL only
B. Add inbound SG rule allowing app server's SG
C. Add outbound SG rule on the database server only
D. Add a route table entry pointing to the app server

14. You have an application deployed in two private subnets across two AZs. It needs outbound internet access. What provides high availability?

A. A single NAT Gateway in one AZ
B. Two NAT Gateways, one in each AZ
C. NAT Instance with Auto Recovery
D. VPC Peering with an internet-enabled VPC

15. A workload requires private connectivity from your VPC to AWS services using interface endpoints. What powers Interface Endpoints?

A. AWS Global Accelerator
B. AWS PrivateLink
C. VPC Peering
D. Transit Gateway

16. Your security team wants to block a specific IP range from accessing a subnet. Which VPC component should be used?

A. Security Group
B. NAT Gateway
C. Network ACL
D. Route Table

17. A VPC has a CIDR 10.0.0.0/16. You want four subnets with equal sizes. Which subnet mask should you use?

A. /18
B. /20
C. /24
D. /22

18. An application in a private subnet needs to connect to DynamoDB without using the public internet. Which is the best solution?

A. NAT Gateway
B. Interface VPC Endpoint
C. VPC Peering
D. Gateway VPC Endpoint

19. Your on-prem network wants to connect to multiple VPCs in AWS using one central connection point. Which service should you use?

A. VPC Peering
B. NAT Gateway
C. AWS Direct Connect Gateway
D. AWS Transit Gateway

20. A security audit shows that someone modified route tables in a VPC. How can you capture future changes for monitoring?

A. Enable VPC Flow Logs
B. Enable CloudTrail for VPC API events
C. Enable GuardDuty
D. Enable Config Rules for EC2

21. A company has two VPCs (VPC-A and VPC-B) connected through VPC Peering. An EC2 instance in VPC-A must access an S3 bucket privately. The same EC2 instance must also access an internal ALB located in VPC-B. However, the EC2 instance cannot reach the ALB. What is the MOST likely reason?

A. Interface endpoints cannot be accessed through VPC peering
B. The ALB subnets do not have a route back to VPC-A
C. S3 Gateway Endpoints block cross-VPC traffic
D. VPC Peering does not support transitive routing

22. Your on-premises datacenter connects to AWS using a Site-to-Site VPN. You now add two more VPCs that also need on-prem access. You want a centralized architecture without creating multiple VPN tunnels. What should you use?

A. VPC Peering among all VPCs
B. AWS Transit Gateway

C. Direct Connect Gateway
D. Multiple NAT Gateways for each VPC

23. A financial application requires private connectivity to Amazon S3 and DynamoDB, no internet exposure, and must log all access attempts. Which combination should you use?

A. NAT Gateway + Flow Logs
B. Internet Gateway + Bucket Policy
C. VPC Endpoints (Gateway + Interface) + CloudTrail
D. Transit Gateway + Security Groups

24. You deployed a NAT Gateway in AZ A. Two private subnets exist: one in AZ A and one in AZ B. The subnet in AZ B is experiencing intermittent internet connectivity. What is the MOST likely cause?

A. NAT Gateway configured incorrectly
B. Route table in AZ B points to NAT Gateway in AZ A
C. NACLs allow only inbound traffic
D. Private subnet's CIDR overlaps with the NAT route

25. A company uses PrivateLink to expose an internal microservice to multiple VPCs owned by partners. One partner reports that DNS resolution works, but traffic never reaches the service. All other partners work fine. What is the likely root cause?

A. The service consumer must use IPv6 endpoints
B. The partner's Security Groups are missing inbound allow rules
C. The partner did not enable the correct DNS hostname type
D. Interface Endpoints require a matching CIDR block

26. You created a VPC with CIDR 10.0.0.0/16. After months, you realize you need more IP addresses for a new application. IPv4 exhaustion is close. What is the MOST cost-effective method?

A. Create another VPC with a non-overlapping CIDR and peer it
B. Delete the VPC and recreate with a larger CIDR
C. Use VPC IP Address Manager (IPAM) to add more CIDRs to the same VPC
D. Use NAT Instances to free IP space

27. Your VPC has a Gateway Endpoint for S3. Instances in the private subnet can reach S3. Suddenly, after adding a new NACL, S3 access stops. What is the MOST likely cause?

A. Gateway Endpoints require DNS support enabled
B. NACL must allow ephemeral ports for return traffic
C. Route table must be updated for the endpoint
D. S3 requires an Interface Endpoint when NACLs are used

28. A 3-tier app (web → app → DB) runs across 3 private subnets. Web tier needs outbound internet for API calls. App tier must remain fully isolated. DB only accepts traffic from app tier. Which architectural configuration is BEST?

A. Attach an IGW to all subnets
B. Add NAT Gateway for web tier; SGs restrict app tier
C. Add NAT to app tier only
D. Use VPC Peering to a public VPC

29. You have set up a Transit Gateway and attached three VPCs. VPC-A should communicate with VPC-B but NOT with VPC-C. What should you configure?

A. NACL blocking VPC-C CIDR
B. Transit Gateway Route Tables with explicit routes
C. Security Group deny rules
D. Modify route tables inside VPC-C only

30. A workload is deployed in VPC with an IPv6 CIDR block. The application must access the internet over IPv6, but not expose inbound connectivity. Which setup is correct?

A. Internet Gateway + NACL deny rules
B. Egress-Only Internet Gateway
C. NAT Gateway with IPv6 enabled
D. Gateway Endpoint with IPv6 routing